Optimizing Training Efficiency Through Data Augmentation: Classical and Advanced Techniques in CNN Models

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Abstract

Recent work has shown that Data Augmentation plays a vital role in enhancing performance of Machine Learning models, especially in field of Computer Vision. This paper investigates the effect of data augmentation on the training efficiency of CNN using fruit image dataset. The study aims to reduce the training time while maintaining or improving the performance of model, it focuses on classical methods like rotation and flipping as well as advance methods like Mixup, Cutmix and RandAugment. Also, this project aims to propose innovative augmentation techniques by combining existing ones. The findings will highlight how effective data augmentation strategies can lead to better generalization and improved model performance in scenarios with limited computational resources. The findings will show how effective data augmentation strategies can significantly enhance model performance and generalization, especially in situations where computational resources are limited.

1. Introduction

In machine learning and deep learning, data augmentation is a technique that is used to artificially increase the diversity of training data without the need to collect more data.

Assisting models in learning adaptive features, common augmentation techniques like rotation, flipping, and noise addition are utilized to enhance generalization. With these, models are able to generalize more effectively , when trained on smaller datasets, and the number of epo chs needed for convergence are decreased.

The problem that we are solving in this project is maintaining model performance while reducing the time spent on training. Large datasets and extended training periods are often required for neural networks to perform efficiently. But computationally, this is costly, particularly when there are resource constraints. We aim to assess how different data augmentation methods can help shorten training times, enhance generalization, and maintain model performance.

2. Related Work

This section provides a brief review of past work that has used data augmentation to improve the performance of image classification or Computer Vision tasks. Classical augmentation techniques such as rotation, flipping, and adding noise have been proven to enhance model performance by increasing the diversity of the training data. For example, Perez and Wang (2017) [3] found that data augmentation significantly improved the accuracy of deep learning models by pro-

viding more varied training examples.

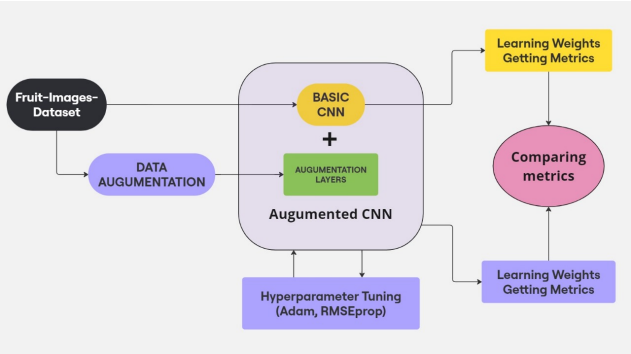


Figure 1. Description of the image.

Advance techniques, such as MixUp [5] and CutMix [4], have come out as effective strategies to reduce overfitting and improve generalization. MixUp linearly combines pairs of images and their labels to create new training examples, which leads model to learn more varied features. CutMix, similarly, combines patches from different images, providing a more diverse set of training examples. These techniques have been shown to lead to excellent results in various computer vision tasks. Moreover, the emergence of automated augmentation strategies like AutoAugment [1] has provided a approach to select the best augmentation method for specific tasks, improving both performance and training efficiency. RandAugment, an automated augmentation method introduced by Cubuk et al. (2020) [2], further builds on previous techniques, instead of searching for optimal augmentation policies (as done in AutoAugment), RandAugment uniformly selects augmentation operations from a predefined set of transformations. It significantly reduces the search space and computational overhead while still achieving competitive results across various datasets. While existing research has made great progress in data augmentation, there is still room to develop innovative methods that combine classical and advanced techniques and improve the existing ones. By exploring new augmentation strategies, this project aims to contribute to the field by improving training efficiency and model robustness, particularly for applications in low-resource environments.

3. Augmentation Strategy Comparison

3.1. Baseline Augmentation Methods and Precomputed Policies

We implemented basic augmentation and random augmentation as baseline methods, alongside precomputed augmentation policies, to compare with our custom AutoAugment policy.

Basic Augmentation involves standard transformations such as resizing, adding noise, adjusting brightness and contrast, rotating, translating, and flipping images. These augmentations are applied uniformly across the dataset to introduce consistent variations.

Random Augmentation selects random transformations at each iteration, varying the augmentations applied in each training cycle without a structured strategy.

Precomputed Policies refer to AutoAugment strategies already optimized for datasets like ImageNet, CIFAR-10, and SVHN. These policies were applied to evaluate how learned augmentation strategies compare to our custom implementation.

These augmentation methods served as benchmarks to assess the effectiveness of our adaptive AutoAugment approach.

3.2. Auto Augmentation: Custom Implementation

The AutoAugment algorithm enhances image datasets through a series of dynamically selected transformations, aiming to improve model performance by introducing diverse variations of the original dataset. This approach operates by applying multiple augmentation policies—such as rotations, color adjustments, and translations—selected based on their historical impact on model performance.

Overview of the Algorithm:

- The algorithm begins with a set of predefined transformations and randomly selects a subset to form the initial augmentation policy.
- For each training epoch, the current policy (comprising three randomly selected augmentations) is applied to each image, resulting in augmented images that maintain key features while introducing variation.
- After training, the resulting accuracy is treated as a

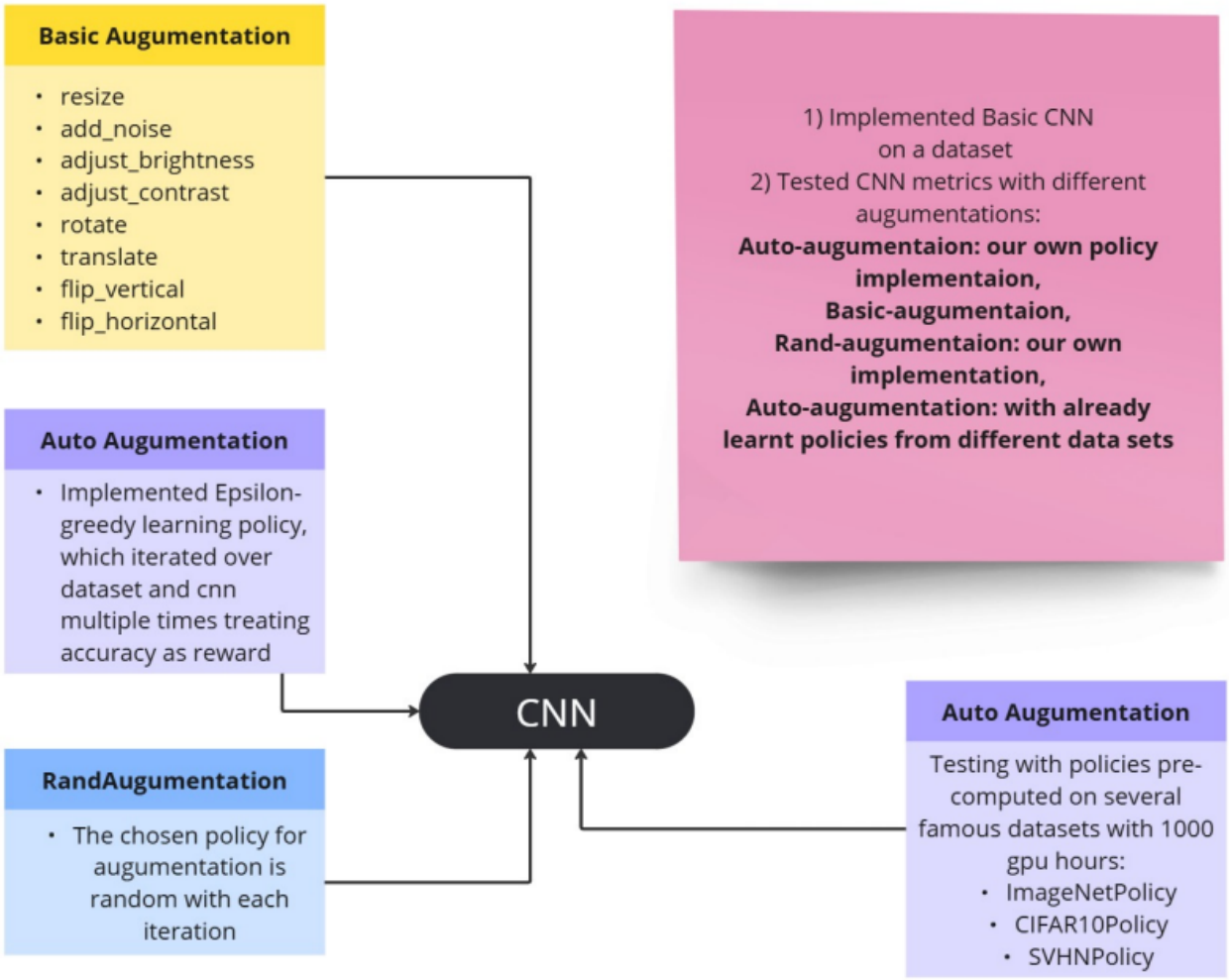


Figure 2. Overview of Data Augmentation Techniques Used in CNN Training

reward in an epsilon-greedy learning policy, which iterates over the dataset and the CNN multiple times. This feedback allows the algorithm to adaptively update its policy for subsequent iterations, rewarding successful augmentations and penalizing less effective ones.

To assess the effectiveness of our implementation, we compared it with established precomputed augmentation policies, specifically ImageNetPolicy, CIFAR10Policy, and SVHNPolicy.

3.3. Conclusion

The results demonstrate that augmentation strategies have a significant impact on model performance. The baseline CNN with no augmentation achieved 70% accuracy. With *basic augmentation*, accuracy improved to 85%, while *random augmentation* further increased it to 90%. Our custom *AutoAugment* policy outperformed these methods, achieving 93% accuracy.

Interestingly, when using precomputed policies, performance varied: the *ImageNet policy* yielded 84% accuracy, the *SVHN policy* 87%, and the *CIFAR-10 policy* 86%. These results highlight the importance of dataset-specific augmentation strategies, with our adaptive *AutoAugment* approach showing the highest gains by dynamically adjusting augmentations to the dataset.

Augmentation	Accuracy
None	70%
Basic	85%
RandAugment	90%
AutoAugment	93%
AutoAugment with ImageNetPolicy	84%
AutoAugment with SVHNPolicy	87%
AutoAugment with CIFAR10Policy	86%

Figure 3. Description of the image.

4. References

[1] E. Cubuk, B. Zoph, D. Mane, and Q. V. Le, "AutoAugment: Learning Augmentation Strategies from Data," arXiv preprint arXiv:1805.09501, 2018. [2] E. Cubuk, B. Zoph, D. Mane, and Q. V. Le, "RandAugment: Practical Data Augmentation with Randomized Policies," arXiv preprint arXiv:1909.13719, 2019. [3] L. Perez and J. Wang, "The Effectiveness of Data Augmentation in Image Classification using Deep Learning," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops, 2017, pp. 1–6. [4] Y. Zhang, T. Wu, G. Zhang, J. Wang, and R. W. H. Lau, "CutMix: Regularization Strategy to Train Strong Classifiers with Localized Patches," Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2019, pp. 6023–6032. [5] S. Zhang, J. C. Song, and M. L. Wong, "MixUp: Beyond Empirical Risk Minimization," International Conference on Learning Representations (ICLR), 2018.

References